Practical 1: Student Marks Analyzer

Problem Definition

Store student data as a list of tuples. Remove duplicate marks, calculate average marks and store result in a dictionary.

Sample Input

```
students = [
    (1, "Amit", [78, 85, 90, 85]),
    (2, "Neha", [88, 92, 79, 88]),
    (3, "Ravi", [70, 75, 80, 75])
]
```

Sample Output

```
{1: ('Amit', 84.33), 2: ('Neha', 86.33), 3: ('Ravi', 75.0)}
```

In [40]:

```
ans = dict()
students = [
    (1, "Amit", [78, 85, 90, 85]),
    (2, "Neha", [88, 92, 79, 88]),
    (3, "Ravi", [70, 75, 80, 75])
]

for i in students:
    temp = []
    for j in i[2]:
        if j not in temp:
            temp.append(j)

    sum = 0
    for h in temp:
        sum+=h
    ans[i[0]] = (i[1],round(sum/len(temp),2))
print(ans)
```

```
{1: ('Amit', 84.33), 2: ('Neha', 86.33), 3: ('Ravi', 75.0)}
```

Practical 2: Inventory Management

Problem Definition

Remove out-of-stock items and calculate total inventory value.

Sample Input

```
inventory = {
    'pen': (10, 50),
    'book': (100, 0),
    'pencil': (5, 100)
}
```

Sample Output

```
{'pen': (10, 50), 'pencil': (5, 100)}
Total Value: 1000
```

In [12]:
```
inventory = {
    'pen': (10, 50),
    'book': (100, 0),
    'pencil': (5, 100)
}
ans = {}
for i,j in inventory.items():
    if j[1] == 0:
        continue
    ans[i] = j
print(ans)
sum = 0
for i,j in ans.items():
    sum+=(j[1]*j[0])
print("Total Value:",sum)
```

```
{'pen': (10, 50), 'pencil': (5, 100)}
Total Value: 1000
```

Practical 3: Email Domain Extractor

Problem Definition

Extract email domains and count occurrences.

Sample Input

```
emails = ['a@gmail.com', 'b@yahoo.com', 'c@gmail.com']
```

Sample Output

```
{'gmail.com': 2, 'yahoo.com': 1}
```

In [23]:
```
emails = ['a@gmail.com', 'b@yahoo.com', 'c@gmail.com']
temp = []

for i in emails:
    temp.append(i.split('@')[1])
ans = {}
```

```
for i in temp:
    ans[i] = temp.count(i)
print(ans)
```

```
{'gmail.com': 2, 'yahoo.com': 1}
```

Practical 4: Shopping Cart Billing

Problem Definition

Calculate item-wise cost and total bill.

Sample Input

```
cart = [('apple',2), ('banana',3)]
prices = {'apple':50, 'banana':10}
```

Sample Output

```
{'apple': 100, 'banana': 30}
Total Bill: 130
```

In [18]:

```
cart = [('apple',2), ('banana',3)]
prices = {'apple':50, 'banana':10}

ans = dict(cart)

for i,j in prices.items():
    ans[i] = ans[i] * j
print(ans)

sum = 0
for i in ans.values():
    sum+=i
print("Total Bill:",sum)
```

```
{'apple': 100, 'banana': 30}
Total Bill: 130
```

Practical 5: Character Position Index

Problem Definition

Store positions of each character in a string.

Sample Input

```
s = "banana"
```

Sample Output

```
{'b': [0], 'a': [1, 3, 5], 'n': [2, 4]}
```

In [24]:
```
s = 'banana'

ans = {}
for i in range(len(s)):
    temp = []
    temp.append(i)
    # j = i+1
    for j in range(i+1,len(s)):
        if s[i] == s[j]:
            temp.append(j)
    if s[i] in ans:
        continue
    ans[s[i]] = temp
print(ans)
```

```
{'b': [0], 'a': [1, 3, 5], 'n': [2, 4]}
```

Practical 6: Survey Data Aggregation

Problem Definition

Count responses city-wise and choice-wise.

Sample Input

```
responses = [
    ('Amit','Rajkot','Yes'),
    ('Neha','Rajkot','No'),
    ('Ravi','Ahmedabad','Yes')
]
```

Sample Output

```
{'Rajkot': {'Yes': 1, 'No': 1}, 'Ahmedabad': {'Yes': 1}}
```

In [39]:
```
responses = [
    ('Amit','Rajkot','Yes'),
    ('Neha','Rajkot','No'),
    ('Ravi','Ahmedabad','Yes')
]

result = {}

for name, city, choice in responses:
    if city not in result:
        result[city] = {}
    if choice not in result[city]:
        result[city][choice] = 0
```

```
    result[city][choice] += 1

print(result)
```

```
{'Rajkot': {'Yes': 1, 'No': 1}, 'Ahmedabad': {'Yes': 1}}
```